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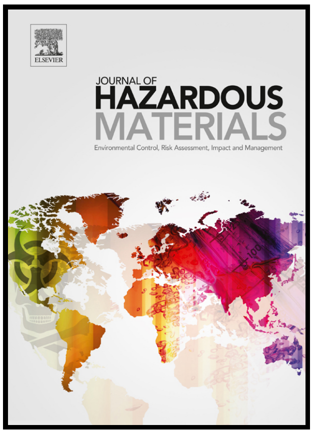

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The Boiling Frog Effect: Global Warming Delays Emotional Impacts of Air Pollution in Warmer Climates

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Abstract

Climate change, air pollution, and extreme weather interact in complex ways that impact emotional states of populations. These dynamics are crucial in effective health planning and risk profiling, however, remain poorly understood in real time contexts, hampering timely responses by governmental agencies. Here we conduct a long-term large-scale investigation of the synthetic and lagged effects of environmental stressors on expressed sentiment, as a proxy of emotional states and subjective wellbeing, derived from over 850 million geotagged tweets using natural language processing across the continental United States from 2016 to 2022. Our spatiotemporal Bayesian hierarchical model, optimized with a distributed lag non-linear algorithm, reveals that combined exposure to air pollution and heatwaves produces significant lagged effects on sentiment, with the population in warmer climates showing more gradual emotional responses than those in colder regions. This evidence corroborates the

'boiling frog effect' – a metaphor implying how populations adapt to environmental stressors in ways that delay emotional responses. These findings provide empirical, spatially explicit support for the long-established environmental psychology conjectures including Environmental Stress Theory and Adaptation Level Theory. Our results offer tangible pathways for wellbeing related interventions, climate adaption strategies and public health emergency response systems in the face of increasing global environmental challenges.

Keywords

Heatwaves, emotional impact, air pollution, environmental stressor, human-environment interaction.

Synopsis

This study reveals how heatwaves and air pollution jointly affect emotional wellbeing, advancing understanding of human adaptation to environmental stress.

Introduction

Extreme weather conditions and air pollution exert profound impacts on emotional states and cognitive performance, affecting both individual's wellbeing and community resilience acutely and chronically(Cao et al., 2023; Cianconi et al., 2020; Weilnhammer et al., 2021). The impact of air pollution on emotional states is a pressing issue that intertwines environmental health, social equity, and climate adaptation, characterized by its hidden, widespread, delayed and unequal effects(Zheng et al., 2019). These impacts intensify with prolonged exposure, leading to anxiety, mental distress, depression, and post-traumatic stress disorder (PTSD)(Buoli et al., 2018; Wang et al., 2022; Yang et al., 2021). However, air pollution and extreme weather rarely operate as isolated factors, rather their interactive and synergistic effects are not merely additive. This highlights the urgency to explore these complex interactions, an area that remains under-studied. Understanding these dynamics and their impacts will provide far-reaching implications for health planning, environmental management and risk profiling(Yang et al., 2018).

Environmental Stress Theory posits that external stressors, including air pollution and extreme weather, negatively influence emotional states and wellbeing(Lazarus and Cohen, 1977). In contrast, Adaptation Level Theory emphasizes individual's adaptation to stimuli based on prior experiences, suggesting that people's emotional responses to environmental stressors are relative to their adaptation levels formed by past exposure and psychological adjustments, which can lead to varying emotional responses(Helson, 1964). Therefore, residents across different regions may experience delayed emotional responses due to adaptive coping mechanisms yet remain vulnerable to stress accumulation(Newbury et al., 2021; Taylor et al., 2004). While air pollution is consistently detrimental to health, extreme weather events produce more varied sentiment outcomes. For example, residents in colder regions may initially perceive heatwaves positively as relief from harsh winters, while those in warmer climates may have developed better adaptation strategies to heat extremes, viewing them as opportunities for outdoor activities and social engagement that positively support mental wellbeing(Jay et al., 2021). This nuanced understanding of extreme weather's dual nature (both as risk and potential benefit) is essential for comprehensive health assessment. Moreover, monitoring the delayed effects of air pollution on sentiment is crucial to provide a time window for public health interventions although it remains as a challenging task(Chen et al., 2017). Current scholarship lacks large-scale longitudinal evidence addressing the synergistic and lagged effects of extreme weather and air pollution on emotional states. Specifically, how these effects vary across populations in different climate zones with distinct adaptation mechanisms remains largely unexplored.

Traditional approaches to measuring emotional states typically rely on subjective assessments by running surveys through questionnaires or interviews(O'Connor et al., 2014; Wei et al., 2016). However, these traditional approaches are often limited in scope and time resolution, arguably struggling in their capacity to capture real-time responses to weather events. Social media analysis has recently emerged and offers a powerful alternative for measuring emotional states based on real-time expressed sentiment(Hu et al., 2021; Siqin et al., 2022; Wang et al., 2023). Multiple validation studies have confirmed tweet-based sentiment measures as reliable alternatives to self-reported life satisfaction surveys and a proxy for wellbeing related constructs(Jaidka et al., 2020; Luhmann, 2017; Pellert et al., 2022; Settanni and Marengo, 2015). This approach enables us to timely quantify the silent emotional burden experienced by millions of exposed individuals who are neither hospitalized nor surveyed during extreme weather and pollution events.

To tackle the above knowledge deficits, our study contributes a large-scale long-term investigation of the synergistic and delayed effects of extreme weather events (i.e., heatwaves and extreme rain) and air pollution on expressed sentiment (hereinafter termed as 'sentiment') across 3144 counties in 48 states in the continental US from 2016 to 2022, using 852+ million geotagged tweets at a daily basis. We built a spatiotemporal Bayesian hierarchical model, optimized by a Distributed Lag Non-linear Model (DLNM) to untangle the spatially and temporally lagged effects of air pollution and extreme weather events on public sentiment. We have three research questions to address: *1) How does air pollution alone affect expressed sentiment compared to the combined effects of air pollution and extreme weather? 2) What is the duration of delayed sentiment reactions triggered by air pollution and extreme weather? 3) How do these effects vary across regions (climate zones and states) and over time?* Our methodological approach creates unprecedented opportunities for predicting population-level emotional responses to natural disasters and public health crises. Most significantly this work provides large-scale real-time empirical support for established but previously unquantified conjectures in environmental psychology, including Environmental Stress Theory and Adaptation Level Theory, advancing our understanding of how humans adapt psychologically to changing environmental conditions in the context of accelerating climate change.

Methods

Our study area encompasses the contiguous United States, including 48 states while excluding the District of Columbia. We collected multi-source data on a daily basis during the summer months (May 1 to September 30) from 2016 to 2022 as below. Our analytical framework is shown in Figure 1.

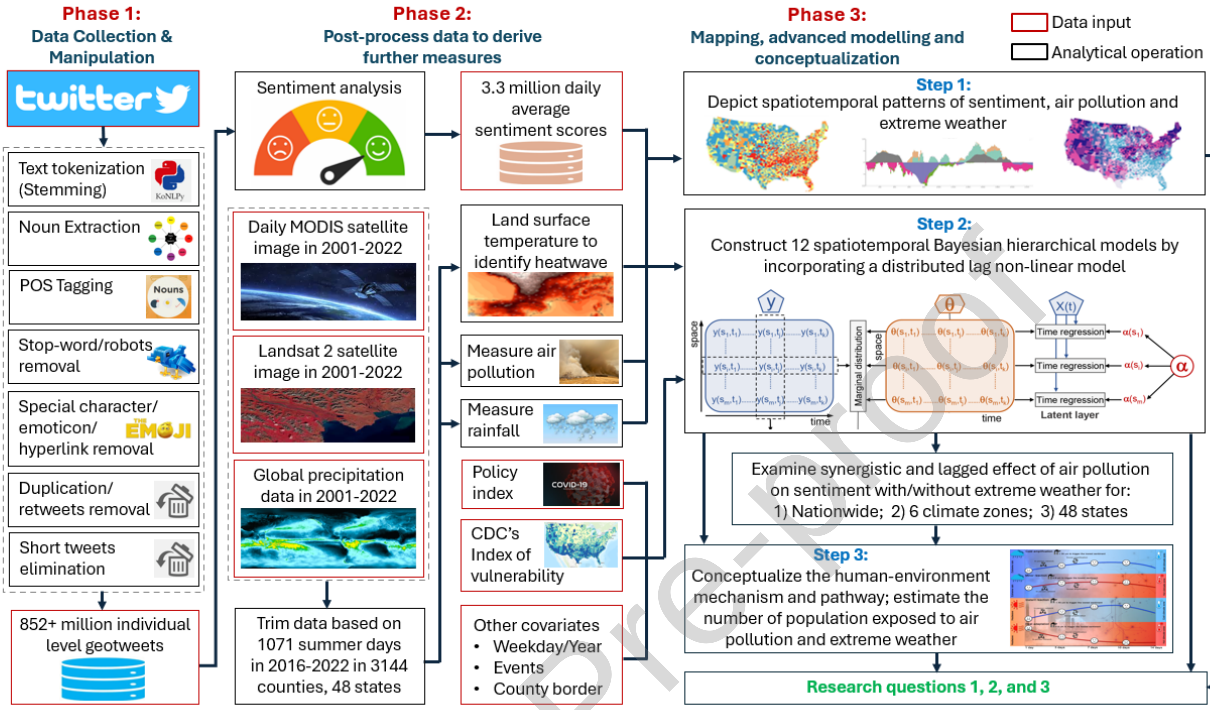

Fig. 1: **Integrated analytical framework: from social media data to climate-emotion dynamics**. In Phase 1, over 852 million individual geotweets were retrieved from the archive tweet dataset from the Center for Geographic Analysis at Harvard University and manipulated through a data processing procedure for sentiment analysis. In Phase 2, a natural language processing model was used to generate sentiment scores at the individual level, which were further aggregated at daily and county level. Heatwaves were defined via a percentile-based method (see *Methods*). Other data, including land surface temperature, air pollution, rainfall, policy index, vulnerability index and other covariates were collected for the study period (May to September, 2016-2022). In Phase 3, a bivariate mapping method was used to examine the spatial and temporal patterns of sentiment, air pollution and heatwave. We then constructed 12 spatiotemporal Bayesian hierarchical models by incorporating a distributed lag non-linear function to monitor the synergistic and lagged effect of air pollution on sentiment with and without extreme weather to provide evidence for conceptualization of the human-environment interactive mechanism.

__Geotweets__: We obtained raw tweet data from the Geotweet Archive v2.0, a vast repository of geographical data developed and stored by the Center for Geographic Analysis at Harvard University since 2012 (Lewis and Kakkar, 2016). The archive is designed to gather tweets with spatial attributes, such as geographical coordinates or place information, if users activate the Global Positioning System (GPS) function in Twitter. During our study period (2016–2022), the platform generated approximately 852 million geotagged tweets, accounting for 1-2% of total tweets. However, this sampling rate does not necessarily indicate bias if geotagging behavior is randomly distributed across the Twitter population. Previous research has found no systematic evidence that users who enable location services differ substantially in sentiment expression patterns compared to non-geotagged users, once accounting for urbanization levels (Jaidka et al., 2020). Our large absolute sample size (852+ million tweets across seven years) provides robust county-level daily aggregates despite the low proportion of geotagged content. Critically, multiple validation studies have demonstrated that Twitter-derived sentiment correlates strongly with gold-standard survey measures of wellbeing and life satisfaction across diverse geographic contexts (Pellert et al., 2022; Luhmann, 2017), suggesting that while Twitter users may not perfectly represent the general population demographically, their expressed sentiment captures meaningful population-level emotional dynamics. Nevertheless, findings should be interpreted as reflecting Twitter users' experiences rather than the entire U.S. population.

We then conducted a series of data preprocessing steps on the geotweet contents. This included removing unrelated website links (such as those starting with "https"), eliminating punctuation marks (e.g., periods, question marks, commas, colons, and ellipses), and removing other key symbols (e.g., brackets, single and double quotes). We also converted all texts to lowercase and remove inflectional endings (e.g., “ly”), returning words to their root or dictionary form using the *word lemmatization* function from the Python package Natural Language Toolkit 3.6.2. It is worth noting that we selected the study period of 2016-2022 rather than utilizing the full geotweet archive (available from 2012) because geotagged tweet volumes and geographic coverage were substantially limited in the 2012-2015 period when Twitter adoption and GPS functionality were still developing. The 2016-2022 timeframe provides more robust and spatially complete daily sentiment estimates across all 3,144 counties analyzed.

Geocoding: We derived the geographic location of geotagged tweets from two primary attributes: geographical coordinates and place metadata. The geographical coordinates attribute provides exact location data, reported by users or their client applications, formatted as longitude followed by latitude in GeoJSON format. The place attribute associates tweets with broader geographic regions, ranging from countries to landmarks, defined by a bounding box. When GPS coordinates are unavailable, we calculated the centroid of this bounding box to approximate the location. To ensure spatial accuracy, we implemented a multi-tiered validation system. Geographic signatures for each tweet—comprising latitude and longitude—were sourced either directly from Twitter’s GPS coordinates or derived from place-based bounding boxes(Hu et al., 2021). A GPS flag system prioritizes direct GPS coordinates over place-based estimates when both are available. Spatial error estimation accounted for precision variations: GPS coordinates were assigned a 10-meter horizontal error margin, while place-based errors were computed as the radius of their bounding boxes. Data reliability was reinforced through rigorous validation. Spatial consistency was verified by cross-referencing coordinates with reported place metadata where possible. We further quantified accuracy using spatial error metrics, ensuring robust geocoding across U.S. administrative levels. This framework supported the processing of approximately 850 million geotagged tweets, balancing scale with precision to uphold analytical integrity.

Heatwave and precipitation: Our study area encompasses the contiguous United States, including 48 states while excluding the District of Columbia. DC was excluded due to its unique administrative status and incompatibility with our county-level spatial modeling framework, which relies on a consistent spatial adjacency structure across county units.
We obtained meteorological data from the North American Land Data Assimilation System Phase 2 (NLDAS-2) covering 48 states in the continental US. It provides data at a spatial resolution of 12.5 km^2 (NASA Goddard Space Flight Center). We used daily mean

temperature data to construct a heatwave index (HI), along with precipitation data for the warm months (May-September) from 2016 to 2022. To define heatwaves, we employed daily mean temperature percentile-based methods, as documented in prior studies, which have shown that relative temperature indices provide optimal predictions for mortality and preterm birth(Anderson and Bell, 2009; Anderson and Bell Michelle, 2011; Kent Shia et al., 2014; Smith et al., 2013). The HI was quantified using a commonly used percentile threshold of 90th, defining heatwave as the periods with daily mean temperatures above the selected percentile threshold of 90^{th} for at least two consecutive days. This percentile threshold was applied to daily temperatures for all days between May 1 and September 30 during the study period (2016-2022). Days identified as within the heatwave were coded as '1' otherwise as '0'. This relative temperature method, which accounted for regional climate differences and local adaptation needs, was particularly well-suited for nationwide research across various climate zones. Moreover, precipitation data were derived from the same NLDAS-2 gridded dataset, with daily total precipitation quantities measured in kg/m^2. Extreme rainfall was defined as precipitation exceeding the 90th percentile of regional precipitation levels (Schär et al., 2016).

Air pollution: Throughout this manuscript, In line with existing studies (e.g., Wei et al. (2020); Gupta et al. (2013)) using aerosol optical depth (AOD) as an air quality proxy refer to "air pollution", we use 'air pollution' to refer specifically to particulate pollution measured through AOD, which quantifies atmospheric aerosol concentrations but does not capture gaseous pollutants such as NO_2, CO, O_3, or SO_2. This terminology is widely understood in the environmental health community when appropriately contextualized. AOD data were obtained from multiple Earth observation datasets via Google Earth Engine for the years 2016-2022, derived from the Moderate Resolution Imaging Spectroradiometer (MODIS) sensors on the Terra and Aqua satellites, provided by the Land Processes Distributed Active Center at the USGS EROS Center(Lyapustin and Wang, 2022). It had a spatial resolution of 1 km and a temporal resolution of 1 day. In the data retrieval process through Google Earth Engine, we used the Science Dataset layer "Optical_Depth_055," which represents the AOD observed in the green light band (0.55 um). Higher AOD values indicate a greater concentration of aerosol particles in the air and, consequently, more severe air pollution. From the perspective of a ground observer, an AOD value below 0.1 corresponds to clear skies, bright sunlight, and maximum visibility. As the AOD increases to 0.5, 1.0, and beyond 3.0, aerosol density increases, eventually obscuring the sun(NASA EarthData). We extracted daily average AOD values at the county level in the US from 2016-2022 by calculating the mean AOD of cells falling within each county polygon.

The time series data from the GEE platform contained null values for some regions and time periods, indicating missing data. To ensure data integrity, we applied both time series and spatial interpolation methods to fill in these gaps. First, time series interpolation was performed using Python's Pandas package. For regions with consecutive missing data for no more than 5 days, cubic spline interpolation was used to estimate the missing values. If any interpolated value was less than 0, it was set to 0. Second, we applied the Gaussian Kriging spatial interpolation to estimate missing values based on nearby regions for the same date. This was done using the *pykrige* package in Python, based on AOD values at the county level.

It is important to note that satellite-derived AOD cannot be retrieved under dense cloud cover or precipitation conditions. Because extreme rainfall events may coincide with cloud contamination and wet deposition processes that physically reduce aerosol concentrations, interpolation during such periods could introduce uncertainty. To evaluate such uncertainty, we compared county-level interpolated MODIS AOD values against collocated ground-based air quality observations from the U.S. Environmental Protection Agency Air Quality System (AQS) monitoring network during the study period. Daily PM2.5 concentrations were aggregated to county level and compared with same-day interpolated AOD using Pearson correlation metrics. Results showed that interpolated AOD retained moderate correspondence with ground observations (correlation coefficient ranging from 0.59 to 0.69), and the estimated lagged effects on sentiment remained qualitatively consistent across datasets, suggesting interpolation does not materially alter the main conclusions (Supplementary Table S1a and S1b).

Other covariates: We further obtained the Social Vulnerability Index (SVI) from the US Center for Disease Control and Prevention(Centers for Disease Control and Prevention and Agency for Toxic Substances and Disease Registry (CDC/ATSDR), 2022) to reflect demographic and socioeconomic status at the county level. The SVI is a composite measure that captures community-level susceptibility to external stresses based on socioeconomic and demographic characteristics. It typically includes factors such as income level, unemployment, education attainment, age structure (e.g., higher proportions of elderly or children), minority status, language barriers, housing conditions, and access to transportation. For example, neighborhoods with higher poverty rates, lower education levels, more elderly residents living alone, and limited vehicle access tend to have higher SVI values, indicating greater vulnerability. In contrast, communities with higher incomes, stable employment, and better housing and transportation resources generally have lower SVI scores.

Additionally, we gathered calendar data indicating holidays and social events at the federal and state levels (with holidays coded as 1 and non-holidays as 0), weekdays (coded as 0) and weekends (coded as 1), weeks (coded as 1 to 52), and years (2016 to 2022). The Policy Stringency Index (PSI), sourced from the Oxford Covid-19 Government Response Tracker(Hale et al., 2021), was included in the models for 2020 and 2021. The PSI is a composite measure that reflects the intensity of government intervention and public health control policies over time. It aggregates multiple indicators such as school and workplace closures, restrictions on public gatherings, stay-at-home orders, travel limitations, and public information campaigns. For example, a low index value corresponds to relatively relaxed conditions (e.g., open businesses, unrestricted mobility), while a high value reflects stricter measures, such as lockdowns, mandatory stay-at-home orders, and widespread closures of schools and workplaces. This index helps capture the broader social and policy environment that may influence public sentiment independently of environmental exposures. The geographic boundary shapefile was retrieved from the US Tiger file(US Census Bureau, 2022) and used to define spatial lags, assigning heavier weights to adjacent counties in the modeling process (see the section on "*Bayesian Hierarchical Modeling*" for details).

Measuring sentiment score: We derived a daily sentiment score to represent population-level emotional states expressed on social media. Each tweet was assigned a sentiment value between 0 and 1 using a pretrained natural language processing model, where values closer to 1 indicate more positive emotion and values closer to 0 indicate more negative emotion. Individual tweet scores were then aggregated to the county–day level to align with environmental exposure data. Conceptually, the sentiment score reflects the overall emotional tone of public expression on a given day. For example, a daily county-level sentiment score of 0.62 suggests generally positive public expression, whereas a score of 0.48 indicates a shift toward more negative sentiment. Changes at the daily scale capture collective emotional responses to real-world conditions. In our dataset, daily sentiment commonly fluctuated within a narrow band (e.g., 0.55–0.65), but noticeable drops were observed following environmental stressors or disruptive events. For instance, in several counties, sentiment scores declined by approximately 0.03–0.06 in the days following elevated air pollution levels, before gradually recovering over the subsequent week. Conversely, holidays and major social events often corresponded with temporary increases in sentiment. These short-term fluctuations provide a meaningful indicator of how collective emotional states evolve in response to changing environmental conditions. Technical details of the language model architecture and training procedure are provided in Chai et al. (2023), as the focus of this study is on the environmental drivers and temporal dynamics of sentiment rather than the sentiment classification method itself.

Bayesian hierarchical modelling: To quantify the relationship between sentiment dynamics and external factors at a granular spatiotemporal scale, we developed 12 spatiotemporal Bayesian hierarchical models (Supplementary Table S2), which is a robust model utilized in the domain(Liu et al., 2023). The dependent variable was the daily county-level sentiment score across the continental U.S. from May 1, 2016, to September 30, 2022. The baseline model accounted for spatial and temporal dependencies using two distinct spatiotemporal random effects, alongside fixed effects for holidays, social events, and a vulnerability index to control for potential confounding factors. Additionally, we divided the research timeframe into three time periods—2016–2019, 2020–2021, and 2022; for 2020–2021 during the COVID-19 pandemic, the Bayesian model included the policy stringency index as the additional confounder to control the fixed effect. The base model formulation is as follows (Liu et al., 2023):

$$S_{i,t} = \alpha + \beta_{c(i)d(t)} + U_{c(i)w(t)} + S_{c(i)} + Vh + Vp_i \tag{1}$$

where $S_{i,t}$ denotes the sentiment score in county i on day t. The first random effect $\beta_{c(i)d(t)}$ captures daily sentiment dependencies on the preceding day through a cyclic first-order random walk (RW1). The second random effect follows a modified Besag-York-Mollié (BYM) structure, combining spatially structured effects $U_{c(i)w(t)}$ to capture regional dependencies in sentiment trends and unstructured county-specific noise $S_{c(i)}$ to address count-level specific characteristics. Additionally, holidays Vh and vulnerability index Vp_i are included as fixed effects. Sentiment scores were modeled using a Gaussian distribution after alternative distributions (Poisson and Binomial) inadequately addressed residual overdispersion in spatiotemporal random effects. To enhance model precision, a Penalized Complexity (PC) prior with $\alpha = 0.01$ was applied to the spatiotemporal random effects. Model parameters were estimated using Integrated Nested Laplace Approximation (INLA) within a Bayesian framework implemented in R (version 4.1.3)(Lindgren and Rue, 2015).

Building on the base model, we further integrated the Distributed Lag Non-linear Models (DLNMs)(Gasparrini et al., 2010) to quantify the non-linear and lagged effects of temperature, air pollution (AOD), and precipitation on sentiment. DLNMs employed natural cubic splines with the capability to represent exposure-lag-response relationships. To examine the lagged effect of air pollution, we set up the lagged duration as 7, 14 and 21 days after the occurrence of air pollution and extreme weather; these are common thresholds have been utilized in existing studies(Liu et al., 2023). After experiments, the 14-day period turned out to be able to capture the lagged effect in majority of states and we then applied it to the models at the climate zone and state levels. The model is expressed as:

$$S_{i,t} = \alpha + \beta_{c(i)d(t)} + U_{c(i)w(t)} + S_{c(i)} + Vh + Vp_i + f.l(heat, d)_i + f.l(aod, d)_i + f.l(prec, d)_i \quad (2)$$

where $f.l(heat, d)_i$,$f.l(aod, d)_i$, $f.l(prec, d)_i$ represent the DLNMs. Model performance is assessed using the Deviance Information Criterion (DIC) and log score metrics. The DIC balances model accuracy against complexity by estimating the effective number of parameters, while the log score evaluates predictive power through leave-one-out cross-validation. For both metrics, lower values indicate superior model performance. If the inclusion of AOD—either as a direct covariate or through the DLNM—does not reduce both the Deviance Information Criterion (DIC) and the log score relative to the baseline model, we interpret this as statistical evidence that air pollution does not provide additional explanatory power for sentiment variation in that region during the study period. In other words, after accounting for spatial, temporal, and other covariates, AOD does not significantly improve model fit.

We then conducted comparative experiments across multiple temporal and spatial scales. Initially, we analyzed county-level data within each state from 2016-2019 and 2022 to test whether heatwaves, air pollution, and precipitation have non-linear lagged effects on sentiment. Data from 2020 to 2021 were reserved for sensitivity analysis to avoid potential lifestyle changes induced by the COVID-19 pandemic. Significant predictors were integrated to assess the joint influences on sentiment within states. Next, the analyses were replicated across six distinct climate zones to explore regional variations. Finally, sentiment responded to combined extreme conditions—such as concurrent air pollution and heatwaves or air pollution and extreme rainfall—were evaluated for each state and climate zone. Extreme rainfall was defined as precipitation exceeding the 90th percentile of daily rainfall within each region over the study period. The compounding model is conducted as follows:

$$S_{i,t} = \alpha + \beta_{c(i)d(t)} + U_{c(i)w(t)} + S_{c(i)} + Vh + Vp_i + f.l(aod, d)_i \times heat \quad (3)$$

$$S_{i,t} = \alpha + \beta_{c(i)d(t)} + U_{c(i)w(t)} + S_{c(i)} + Vh + Vp_i + f.l(aod, d)_i \times prec \quad (4)$$

After determining the optimal models, we computed the suppress risk (SR) of sentiment changes during severe air pollution episodes relative to no air pollution baseline conditions (AOD = 0.1) as follows:

$$SR = \frac{P(R|E)}{P(R|\neg E)} \quad (5)$$

Here, $P(R|E)$is the posterior ratio of sentiment score under exposure and $P(R|\neg E)$ represents the prior ratio under baseline conditions. Thus, SR > 1 indicates sentiment improvement, while SR < 1 indicates sentiment deterioration. For each experiment, we further interpreted: (1) the minimum sentiment score triggered by air pollution; (2) the AOD value corresponding to this minimum sentiment score; and (3) the temporal window of sentiment reduction at the identified critical AOD.

It is worth noting that in addition to extreme-event indicators (heatwaves and extreme rainfall), baseline meteorological conditions were accounted for in the modelling framework through daily temperature and precipitation variables. This design allows us to separate the gradual emotional influence of normal weather variability from the additional psychological stress induced by threshold-based extreme events. Our primary focus is therefore not on the standalone effect of ordinary meteorological fluctuations, but on how extreme conditions modify or amplify the relationship between air pollution and expressed sentiment. This approach enables clearer interpretation of whether extreme weather introduces nonlinear or delayed emotional responses beyond those associated with typical daily weather conditions.

Robustness and sensitivity analysis: We compared the effect of AOD on sentiment in two scenarios: a DLNM model incorporating only AOD, and a DLNM model that jointly considers heatwaves, AOD, and precipitation. We compared the effect of AOD on sentiment under two scenarios: (1) a model including only the DLNM term for AOD, and (2) a model including separate DLNM terms for air pollution, heatwaves (temperature), and precipitation simultaneously. If the results indicate that including multiple DLNM terms do not significantly alter the impact of AOD on sentiment, it confirms the robustness of our findings. For sensitivity analysis, we evaluated model performance in each state by exploring whether incorporating DLNM enhances explanatory power compared to directly including the variable itself. This aims to determine whether the variables (e.g., air pollution and extreme weather) exhibit a non-linear lagged effect on sentiment scores. In addition, we compared the effects of both direct exposure and DLNMs for the three environmental indicators on sentiment across three periods—pre-pandemic (2016–2019), during the pandemic (2020–2021), and post-pandemic (2022). We found that the lagged effects of AOD consistently explained sentiment variations in both the pre- and post-pandemic periods.

Results

Evolving landscape of sentiment, air pollution and heatwaves

To better understand the potential interrelationships among heatwaves, air pollution and sentiment across the U.S., we first outline their spatial and temporal patterns (Fig. 2). Higher levels of air pollution and heatwaves exhibit clear spatial clustering, with high occurrences along the northwest coast and the northern U.S./Canada border, particularly in the northern states (e.g., Oregon, Washington, Montana, and North Dakota) from 2017 to 2022, and in the southern states (e.g., Texas and Florida) in 2016, 2019, and 2020 (Fig 2A). In contrast, public sentiment exhibits a sporadic spatial pattern without a clear overarching trend over the entire study period (Fig. 2B). Temporally, heatwave days were most frequent in July and August, peaking in mid-July across 2016–2022, except for 2021 (Figure 2C). However, the daily average sentiment score (blue solid line in Figure 2C) and air pollution levels (green dashed line in Figure 2C) show fluctuations rather than seasonal patterns as reported in previous studies(Buchholz et al., 2022; Kuerban et al., 2020; Xu et al., 2023). It is worth noting that although individual tweet sentiment scores range from −1 to +1, Figure 2C shows daily averages aggregated at the regional level. Due to the predominance of neutral-to-positive expressions in large-scale social media data, the observed daily mean values cluster within a positive range (approximately 0.5–0.7). The quintiles shown in Figure 2B were constructed based on the empirical distribution of these daily average scores, rather than the theoretical −1 to +1 scale. Thus, the lowest quintile represents relatively lower sentiment days within the dataset, not necessarily negative absolute sentiment values.

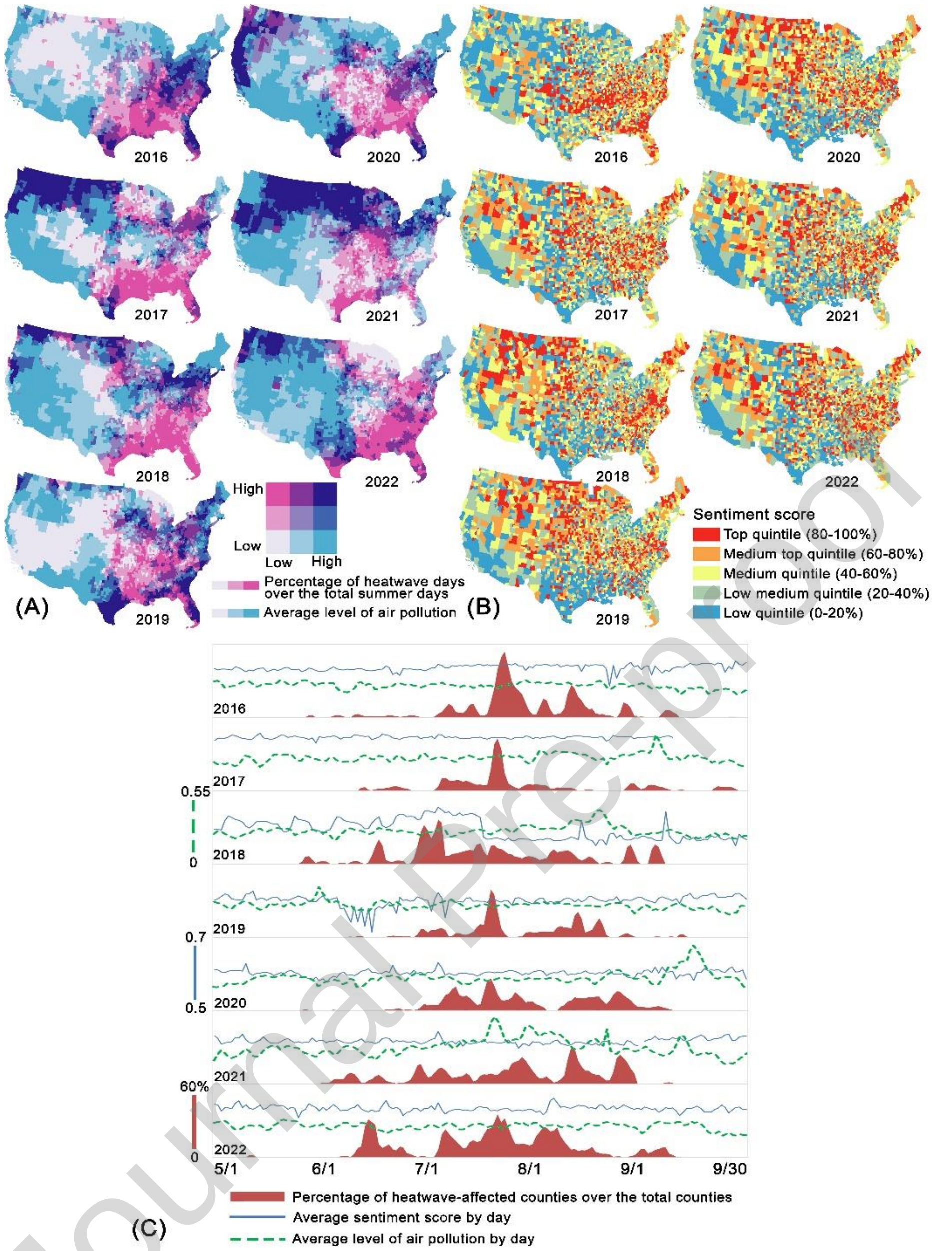

Fig. 2: Spatial and temporal patterns of air pollution, sentiment and heatwaves. (**A**) Bivariate maps simultaneously display two environmental stressors: the percentage of heatwave days over total summer days (pink color ramp, increasing from light to dark pink) and average aerosol optical depth as a measure of particulate air pollution (blue color ramp, increasing from light to dark blue). Dark purple areas indicate regions experiencing both high air pollution and extensive heatwave exposure, representing populations at highest risk for combined exposure effects. Light colors indicate low values on both dimensions, while intermediate purple shades show moderate levels of both stressors. (**B**) Choropleth maps depict public sentiment levels, ranging from -1 (most negative) to +1 (most positive), categorized into five quintiles. (**C**) Line and ridge graphs display temporal trends, showing heatwave occurrences (red areas), daily average sentiment scores (blue line), and daily average air pollution levels (green dashed line) across five summer months from 2016 to 2022. The Y axis here represents the daily mean Aerosol Optical Depth (AOD). AOD is a dimensionless (unitless) measure of atmospheric aerosol concentration, where higher values indicate greater levels of air pollution.

Synergistic effect of air pollution and extreme weather on sentiment

To investigate the interaction between air pollution, extreme weather, and sentiment, we begin by constructing a basic spatiotemporal Bayesian hierarchical model using key covariates, including the vulnerability index, holidays and social events, weekdays, policy stringency index during the COVID-19 pandemic(Hale et al., 2021). We then sequentially introduce air pollution, heatwaves, and extreme rain as individual, independent factors, both with and without spatiotemporal lagged effects (see *Methods*). The results indicate that incorporating spatiotemporal lagged effects for these factors enhances model performance at various levels (Supplementary Table S3-S5 and Figure S1-S4). The next step is to involve the combined exposure to air pollution and extreme weather (heatwaves and extreme rain) with a spatiotemporal lagged effect, yielding the results presented below.

Suppress risk (SR) measures the extent to which sentiment changes from the normal baseline, namely, SR = 1 reflecting a neutral status in sentiment (SR >1 reflecting an increase in sentiment while SR <1 reflecting a decline in sentiment) when exposed to air pollution, with or without extreme weather (Figure 3A). A lower SR indicates a more significant drop in sentiment, with the most severe decline occurring at the highest level of air pollution (black dot in Figure 3B). At the national level, the level of air pollution that begins to trigger a decline in sentiment and the level of air pollution when sentiment drops to the lowest are 0.18 (detailed statistics provided in Supplementary Note S1) and 2.1 (Figure 3B), respectively, during heatwaves, compared to 0.15 and 1.65 in the absence of heatwaves. Given that an AOD below 0.1 signifies clear skies with maximum visibility, around 0.4 indicates hazy conditions, and above 2.5 represents severe air pollution with reduced visibility (Kessner et al., 2013; Zhang et al., 2020), our findings suggest that air pollution does not need to reach the hazy condition to elicit a sentimental response.

A similar pattern is observed with extreme rain, where air pollution must reach a higher threshold to impact sentiment compared to normal rain. Specifically, sentiment declines at an AOD level of 0.15 (1.85 when sentiment drops to the lowest) during extreme rain, compared to 0.15 (1.6) during normal rain (Figure 3B). This suggests that the psychological stress induced by extreme temperatures and heavy rainfall may overshadow the effects of air pollution until pollution levels surpass a critical threshold, intensifying physical stress responses and leading to a delayed emotional reaction(Hswen et al., 2019; Li et al., 2015).

The impact of air pollution on sentiment varies by climate zone. During heatwaves, air pollution has the most pronounced effect on sentiment in the cold mixed humid zone, with a minimal SR of 0.90 [95% credible intervals (CrI): 0.70, 1.16] (Figure 3A), followed by the cold and very cold zone (0.93 [0.90, 0.97]) and the cold dry zone (0.99 [0.79, 1.15]) during heatwaves. In contrast, the corresponding SR values under normal temperatures are 0.74 [0.71, 0.77], 0.89 [0.88, 0.91], and 0.98 [0.96, 1.01], respectively. These results suggest that in colder regions, air pollution has a greater impact on sentiment under normal temperatures than during heatwaves. However, the opposite pattern is observed with extreme rain, that is, air pollution more significantly lowers sentiment during extreme rain than under normal rainfall conditions. Extreme heavy rain has been observed to heighten feelings of vulnerability and distress (Hamada et al., 2015); the combined exposure to air pollution and extreme heavy double the adverse effects on sentiment. Residents in cold regions may be psychologically adapted to frequent rain and its associated pollution dynamics but perceiving extreme rain as an environmental stressor. This resilience could mitigate the immediate emotional impact of combined air pollution and increased rain. Previous studies suggest that in cold climates where well-established coping mechanisms exist for dealing with adverse weather, residents may exhibit a delayed negative response to rising pollutant levels during extreme rain(Shaposhnikov et al., 2014).

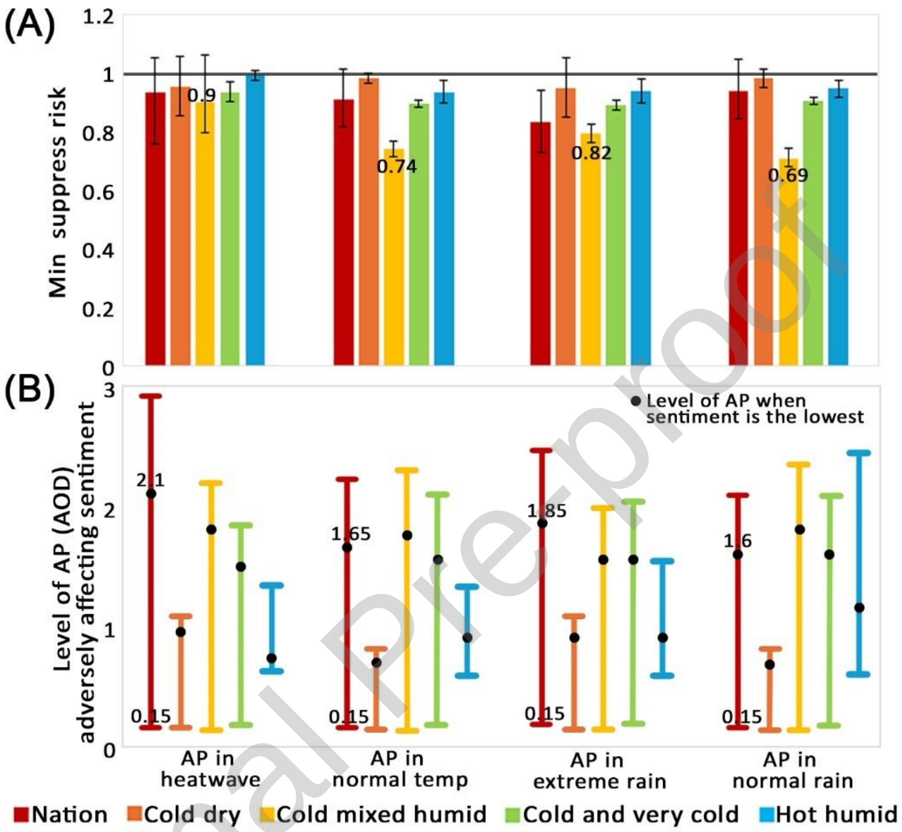

Fig 3: Synergistic effect of air pollution on sentiment at both national and climate-zone levels, with and without exposure to extreme weather events. (**A**) The Y axis displays the minimal suppress risk (SR), indicating the point at which the combined exposure to air pollution and extreme weather has the most significant impact on sentiment. A minimal SR below 1 signifies a negative effect on sentiment. (**B**) The Y axis shows the combined exposure to extreme weather and air pollution level at which sentiment is most affected, represented by a black dot, with 95% CI indicated by vertical bars. X axis of both A and B shows the four scenarios—air pollution in heatwave, in normal temperature, in extreme rain and in normal rain. Four climate zones are presented here due to that air pollution does not significantly affect sentiment in other climate zones. The delineation of climate zones is provided in Supplementary Figure S5.

Lagged effect of air pollution and extreme weather on sentiment

Our Bayesian hierarchical model examines the lagged effect within 14 days after the occurrence of air pollution and extreme weather, a common threshold that has been utilized in existing studies (see *Methods*). At the national level, air pollution begins to trigger a decline in sentiment on Day 4 during heatwaves, compared to Day 3 under normal temperatures (Figure 4A and 4C). By Day 7, air pollution causes sentiment to drop to its lowest point in both conditions. Regarding rainfall, air pollution leads to an immediate (Day 0) decline in sentiment following extreme rain, reaching its lowest point on Day 2. In contrast, under normal rain conditions, sentiment starts to decline on Day 5 and reaches its lowest on Day 8 (Figure 4B and 4C). At the climate-zone level, air pollution triggers a sentimental decline on Day 5 and reaches its lowest point on Day 7 in the hot-humid zone when exposed to heatwaves. In contrast, in the cold mixed-humid and cold/very cold zones, sentiment drops immediately following a heatwave. This suggests that individuals in hotter regions, who are more acclimatized to high temperatures, exhibit greater resilience and a delayed response to the combined effects of air pollution and heatwaves (Figure 4A and 4C).

Regarding rainfall, air pollution triggers an immediate decline in sentiment following extreme rain in the hot-humid and cold-dry zones. However, in the cold mixed-humid and cold/very cold zones, air pollution exhibits a significantly delayed effect, influencing sentiment up to the end of the 14-day period (Figure 4B and 4C). This highlights the complex lagged effects of air pollution and extreme weather on sentiment, particularly across different climatic regions(Zhao et al., 2022). Individuals in hotter climates may experience slower sentiment shifts in response to air pollution, potentially due to physiological adaptation, psychological resilience, and the nature of extreme weather events, which mitigate immediate emotional reactions(Fang et al., 2017; Li et al., 2020; Noback et al., 2011).

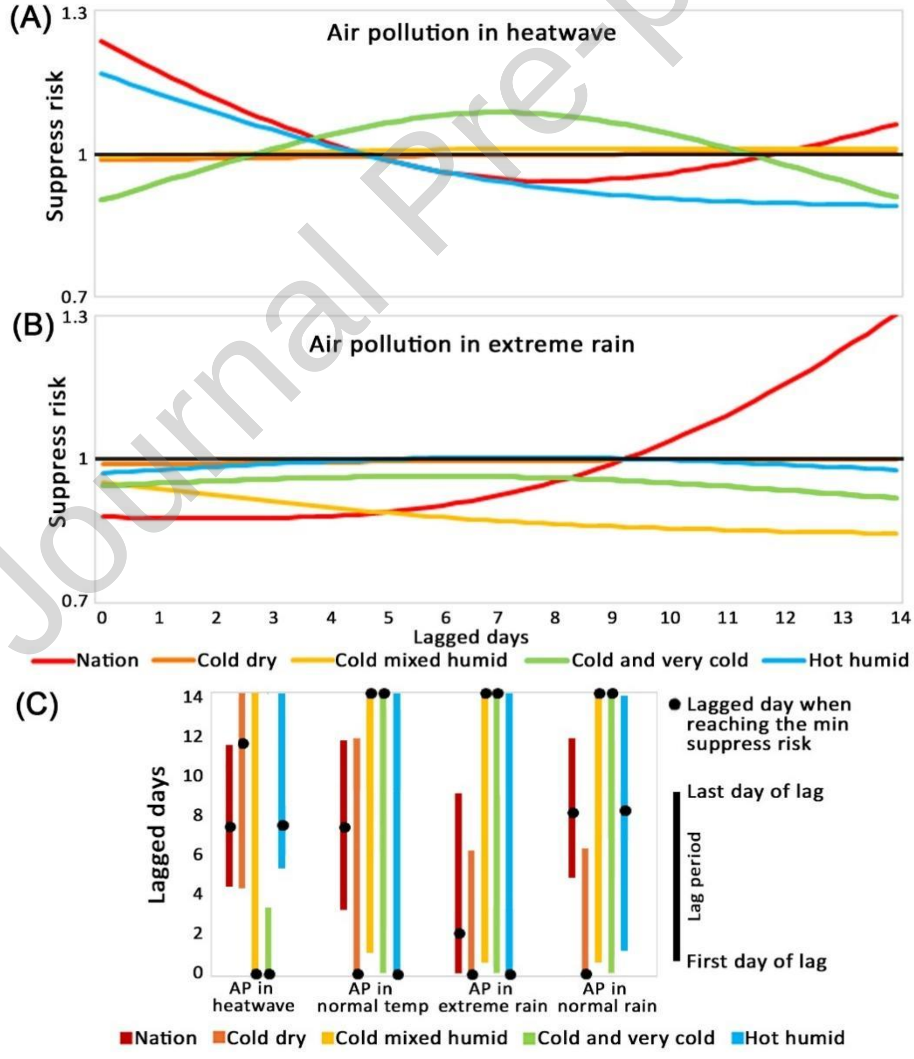

Fig. 4. Lagged effects of air pollution on sentiment with and without exposure to extreme weather at the national and climate-zone levels. (A) and **(B)** illustrate the overall relationship between air pollution, extreme weather (heatwaves and extreme rain), and sentiment, highlighting the day air pollution begins to reduce sentiment. This is indicated by the intersection of each curve with the black horizontal line at SR = 1. A minimal SR < 1 signifies a decline in sentiment, while the lowest point of the minimal SR marks the most significant sentiment decline, corresponding to a specific lagged day. **(C)** depicts the lag duration during which air pollution influences sentiment. The length of the vertical bar represents this duration, while the black dot indicates the exact day when sentiment declines most significantly due to air pollution. The lagged effects of air pollution as a single factor, air pollution in the normal temperature and air pollution in the normal rain are provided in Supplementary Figure S6.

Local variation across 48 states

The interrelationships among sentiment, air pollution, and extreme weather vary across the 48 continental U.S. states but exhibit common patterns in the northern and southern regions. In tropical southern states, air pollution has a less pronounced effect on sentiment during heatwaves compared to normal temperatures, as indicated by the higher red bars (heatwaves) than blue bars (normal temperature) in Figure 5A. This pattern is evident in Texas (AP level = 2.45 during heatwaves vs. 1.35 in normal temperatures), Florida (0.5 vs. 1.3), Louisiana (0.48 vs. 0.88), and New Mexico (0.54 vs. 0.92). Conversely, in northern states, air pollution has a stronger impact on sentiment during heatwaves than under normal conditions. For instance, North Dakota (1.6 vs. 2.05), Wyoming (0.95 vs. 1.1), Minnesota (1.65 vs. 2.55), Wisconsin (2.15 vs. 2.15), New York (2.7 vs. 1.6), Vermont (1.3 vs. 1.35), New Hampshire (1.5 vs. 2.25), and Maine (1.35 vs. 1.9) follow this trend. Among the 48 states, 23% experience the most significant sentiment decline at an AP level of 1.2–1.6, while another 23% show this effect at an AP level of 0.4–0.8. Additionally, 9% of states require an AP level above 2.0 to trigger sentiment decline, with 3% falling in the 2.0–2.4 range and 6% in the 2.4–2.8 range. States with the highest AP thresholds include Arizona, Colorado, Delaware, Florida, and Georgia (Figure 5B).

Furthermore, the lagged effect in southern states such as Texas, Florida, Louisiana, Mississippi, Oklahoma, and Georgia display a downward pattern (red line in Figure 5C), interacting with the black dashed line (indicating minimal SR at 1) several days after air pollution occurs. This reflects the delayed impact of air pollution on sentiment. Initially, air pollution has no immediate adverse effect on sentiment, but after a lagged period—6 days in Texas, 3 days in Florida, 6 days in Louisiana, 2 days in Mississippi, and 3 days in Oklahoma—air pollution begins to negatively affect sentiment. These lagged effects persist throughout the 14-day period. In contrast, states in the central north (e.g., North Dakota, Minnesota, Wisconsin) and the northeast (e.g., New York, Vermont, New Hampshire, Massachusetts, Maine) show an upward pattern in the lagged effect of air pollution on sentiment during heatwaves. At the start of a heatwave, air pollution immediately influences sentiment, causing a sharp drop, which gradually returns to normal levels (i.e., the black dashed baseline when SR = 1).

It is worth noting that while strong temporal autocorrelation in environmental exposures theoretically poses challenges for identifying precise lag-specific effects, our

Bayesian hierarchical framework explicitly addresses this through temporal random effects, smoothing splines, and penalized complexity priors. Our substantive conclusions focus on robust, well-identified features—the presence of lagged effects, the approximate timing of response onset, and qualitative regional differences (immediate vs. delayed patterns)—rather than precise day-specific effect magnitudes. The consistency of lag patterns across 48 independently estimated state models, combined with strong model performance metrics, provides confidence that our findings reflect genuine environmental-sentiment relationships rather than statistical artifacts of autocorrelation.

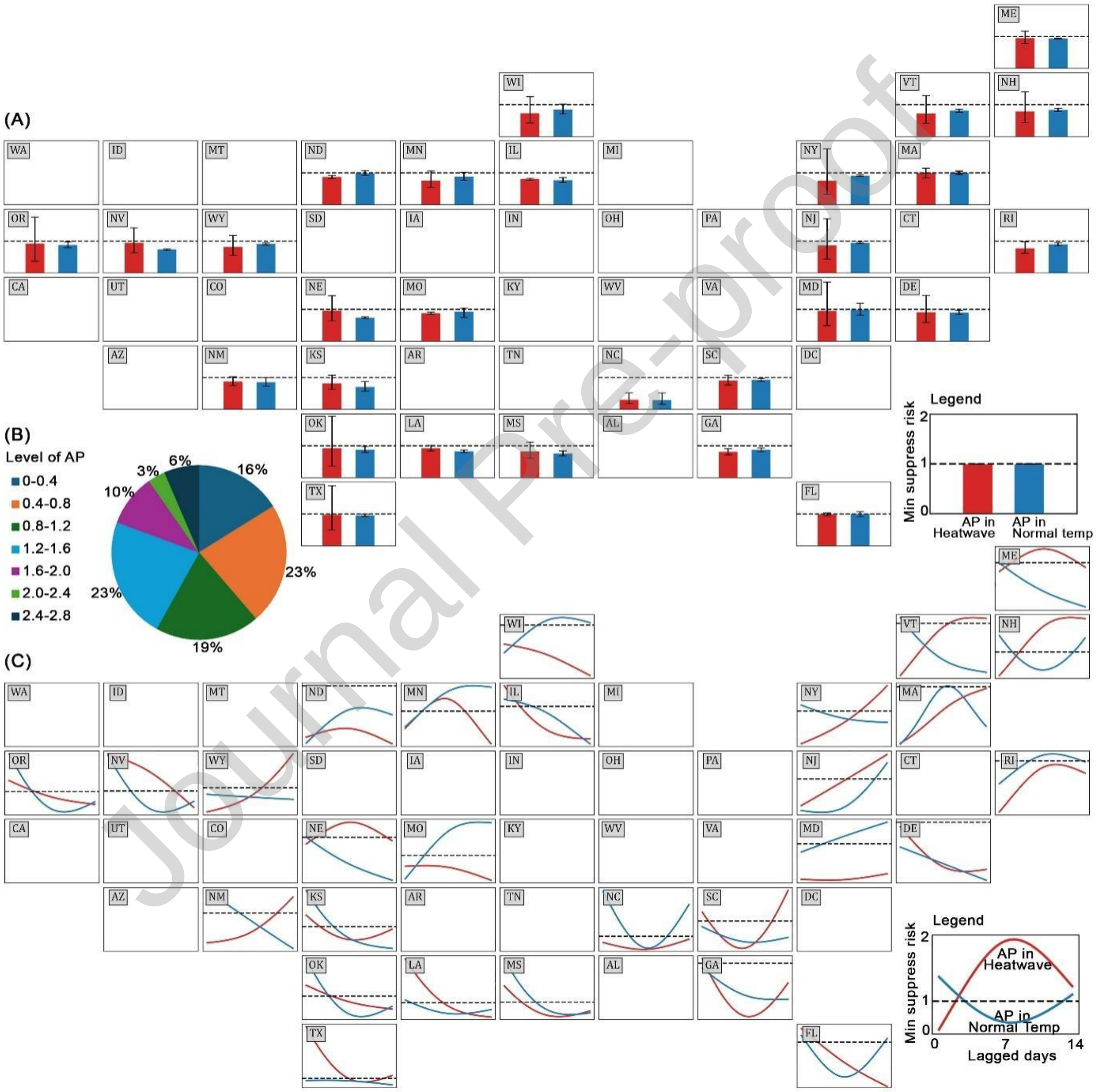

Figure 5. Local variation in the interrelationships among sentiment, air pollution, and heatwaves across the 48 continental U.S. states. **(A)** displays bar charts for each state showing the minimal SR of how air pollution affects sentiment, with exposure to heatwaves (red bars) and without exposure to heatwaves (blue bars). **(B)** shows the proportion of states, out of the total 48, where a specific range

of air pollution begins to trigger a decline in sentiment. **(C)** presents line charts for each state, illustrating the relationship between the minimal SR (Y-axis) and lagged days (X-axis), with the red line representing air pollution during heatwaves and the blue line during normal temperatures. Results for extreme rain and normal rain are provided in Supplementary Figure S7-S10. Blank area in states show no significant sentiment changes linked to air pollution.

Key findings

While the adverse effects of air pollution on emotional states are well documented, our findings reveal a more complex interaction between air pollution, extreme weather, and sentiment dynamics, challenging the one-size-fits-all theory to explain psychological responses across all regions. Conventional models, such as Environmental Stress Theory, assume a uniform relationship between air pollution and psychological distress, but our findings suggest that extreme weather events can alter and even delay these effects, with regional variations playing a significant role. Specifically, air pollution at the national level can intensify during extreme weather events like heatwaves or heavy rainfall, reaching levels severe enough to dampen public sentiment. However, its impact varies by temperature context: in colder regions, pollution's effect on mood is more pronounced under normal temperatures than during heatwaves, whereas during extreme rain, the pattern reverses. The timing of these emotional shifts also differs. During heatwaves, a decline in sentiment begins on the fourth day of pollution exposure, hitting its lowest point by day seven. In contrast, extreme rain triggers an immediate drop in sentiment, with the lowest mood recorded as early as the second day. Regional climate differences further shape these trends. In hotter climates—such as southern states like Texas, Florida, Louisiana, and New Mexico—air pollution's influence on sentiment emerges later, starting on the fifth day of a heatwave. Conversely, in colder northern regions and New England, public mood declines sharply right at the onset of a heatwave, with no delayed effect.

Contribution to the literature

Our findings advance the theoretical foundation of environmental psychology by enriching and extending the long-standing conjectures—such as the classic Environmental Stress Theory(Lazarus and Cohen, 1977) and Adaptation Level Theory—through real-time place-based evidence in a long term. Given the nature of tweet-based sentiment metrics derived from social media, to explain these nuanced sentiment dynamics, we integrate the Social Amplification of Risk Framework (SARF)(Kasperson et al., 1988) with Adaptation Level Theory (ALT) (Helson, 1964) to conceptualize the human-environment mechanism and reveal the pathway through which air pollution and extreme weather shape the sentiment landscape (Figure 6). SARF breaks down the evolution of sentiment into four distinct stages, which can interact with ALT. ALT posits that individuals adapt to their baseline environmental conditions, influencing their perception of new stimuli. In the first phase, individuals begin engaging in face-to-face conversations, influenced by air pollution and weather events. People often increase outdoor activities and social interactions on sunny, hot days, which may boost their sentiment. As the process moves into the second phase, individuals increasingly turn to social media platforms to digitally express their emotions and attitudes. This expression can quickly spread across networks, creating an amplification effect in the third and fourth phases. If extreme weather overshadows the detrimental effects of air pollution, individuals may be less inclined to express immediate concern on social media, leading to a delayed reaction as the issues eventually become more prominent. Furthermore, ALT suggests that people in hot and humid regions, such as Florida and Texas, may have a higher adaptation threshold to heat, making them less sensitive to heatwaves compared to those in cooler regions. This adds another layer of complexity to the explanation of the air pollution-sentiment nexus.

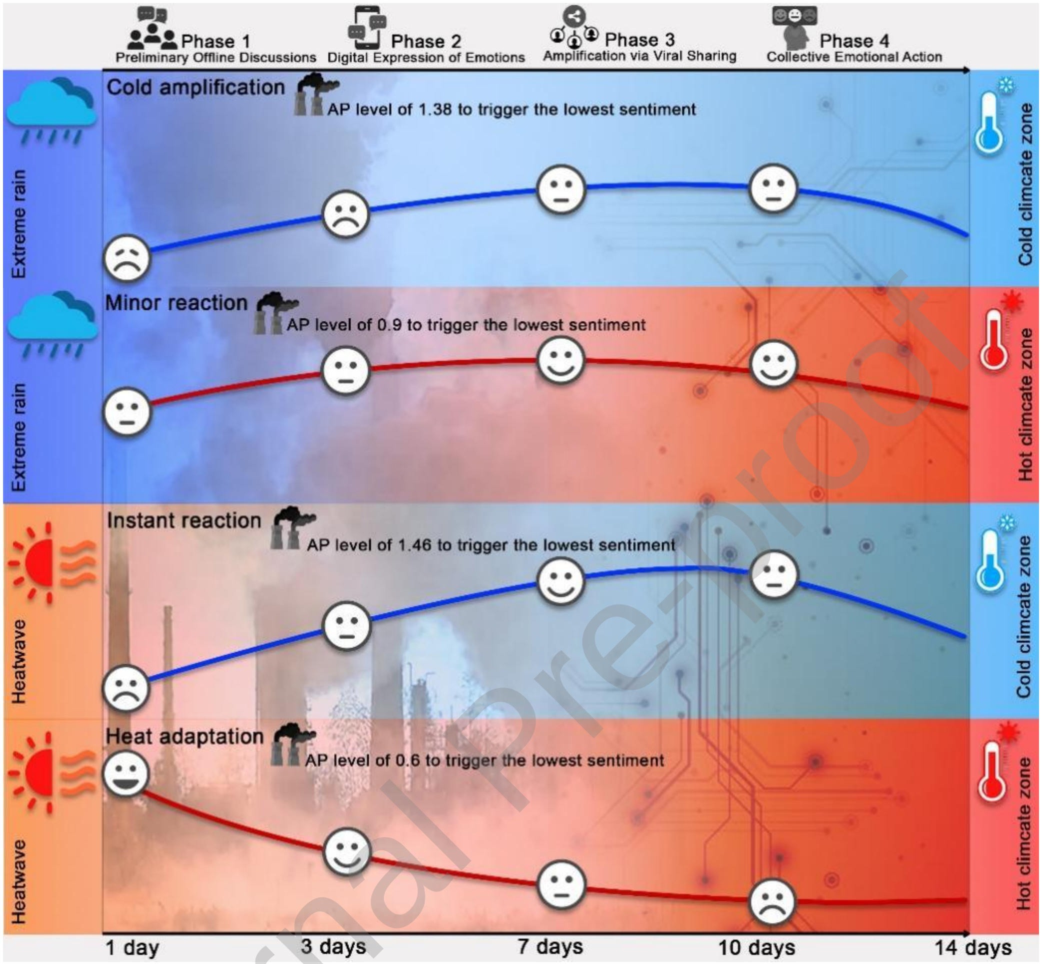

Figure 6. Conceptualization of sentiment-air pollution nexus mediated through extreme weather.

Policy implications

Our findings provide far-reaching policy implications for place-based health planning and community intervention for emotional states and wellbeing—which are critically important to 30-32 million people affected by air pollution and extreme weather in the US (Supplementary Table S6). Such numbers will further increase in the future. First, our results bear out the suggestion for climatic change researchers and air quality modelers, to prioritize longitudinal studies that examine the long-term psychological effects of air pollution and extreme weather across diverse climatic zones. Such studies are crucial for understanding how adaptation levels vary among different populations. Second, policy makers and health planning authorities should consider creating community-targeted and place-based health interventions that account for the unique socio-environmental contexts of different regions. Programs that build community resilience through public health infrastructure, education, and social support networks are critical. Recognizing the lagged sentiment responses observed in colder climates leading to longer reaction times can shape timely public health advisories and response strategies—especially during extreme weather events when pollution levels spike. Third, community engagement can implement communication strategies that actively inform the public about air quality and health implications, particularly during extreme rain or heat events. Regular workshops on coping strategies and community resilience can enhance public preparedness through supportive social networks.

Limitations

Importantly, our findings should be interpreted in light of several study limitations. First, although social media sentiment analysis (e.g., expressions of happiness) has been linked to emotional states, our county-level metrics derived from Twitter language processing may include inaccuracies, as online platforms do not fully represent broader populations in different regions (Huang et al., 2024). Prior research highlights that Twitter users often differ demographically and behaviorally from the general public(Huang et al., 2022), potentially limiting the generalizability of our results. Future studies can derive a climate-specific or pollution-integrated sentiment score that can be readily applied by end users. Furthermore, the demographic characteristics of Twitter users—who tend to be younger, more educated, and more urban than the general population—may introduce selection bias in our sentiment measures. This is particularly relevant for interpreting our regional heterogeneity findings, as the socio-demographic composition of Twitter users varies between northern and southern U.S. regions. The observed 'boiling frog effect' in warmer climates, where populations show slower sentiment decline during combined air pollution and heatwave exposure, may reflect not only climatic adaptation but also differences in who is represented in our Twitter sample across regions. Urban Twitter users in southern states may have greater access to climate-controlled environments and adaptation infrastructure than rural populations, potentially moderating their expressed emotional responses. Future research should conduct stratified analyses comparing urban versus rural populations to disentangle climatic adaptation mechanisms from demographic selection effects. Additionally, integrating Twitter sentiment data with traditional survey methods across diverse demographic groups would strengthen the generalizability of findings and clarify whether the observed regional patterns reflect genuine climate-based adaptation or sampling artifacts."

Second, environmental variables such as temperature, precipitation, and air pollution, measured via satellite remote sensing, are susceptible to biases like atmospheric interference (e.g., cloud cover) and constraints in geographic or time-related precision. Furthermore, satellite-based AOD retrieval is not available under cloudy or rainy conditions, requiring interpolation of missing values. Although we applied both temporal spline interpolation and spatial Gaussian Kriging to minimize discontinuities, extreme rainfall may reduce aerosol concentrations through wet deposition. In such cases, interpolation could partially attenuate the true washout effect and slightly overestimate air pollution levels. However, validation against in-situ monitoring data and sensitivity analyses excluding extreme rainfall periods indicate that our core findings remain stable. Therefore, any potential bias is unlikely to overturn the main conclusions, though future research integrating ground-based air quality measurements at finer temporal resolution would further strengthen inference.

Third, our approach to filling missing AOD values through temporal spline interpolation and spatial kriging may smooth extreme pollution episodes, potentially leading to conservative estimates of sentiment impacts during severe air quality events. Both cubic spline interpolation and Gaussian Kriging, by design, produce smoothed predictions that may attenuate peak AOD values during acute pollution episodes such as wildfire smoke plumes or industrial pollution events. This smoothing could underestimate the true magnitude of sentiment decline during severe pollution episodes and potentially miss shorter-duration but high-intensity events that trigger acute emotional responses. Nevertheless, future studies incorporating multiple satellite products or ground-based monitoring networks could improve precision in capturing pollution extremes and provide more accurate estimates of sentiment impacts during the most severe air quality events.

Fourth, the regional differences we observe in immediate versus delayed sentiment responses may be partially attributable to variation in pollution sources rather than solely reflecting temperature-based adaptation mechanisms. Summer AOD elevations in northern and western states are frequently driven by wildfire smoke, which produces acute, visible pollution episodes that may trigger immediate emotional responses due to their salience and associated health warnings. Conversely, southern states often experience more chronic pollution from industrial sources, vehicle emissions, and photochemical smog, which accumulate gradually and may produce more delayed psychological responses. Wildfire smoke also carries distinct health risks and social meanings (natural disaster, evacuation threats) compared to urban air pollution, potentially eliciting different emotional reactions independent of climate adaptation. Future research should disaggregate AOD by pollution source type (wildfire vs. anthropogenic) to disentangle the relative contributions of pollution characteristics versus climate-based adaptation in shaping temporal patterns of sentiment response.

Finally, the delayed sentiment response in warmer climates may reflect behavioral avoidance strategies rather than purely psychological adaptation mechanisms (Ahmed, 2010). Residents in southern states likely have greater access to air conditioning and climate-controlled environments, enabling them to remain indoors during combined heat and pollution events. This behavioral buffering could delay direct exposure to outdoor air quality, postponing the physiological discomfort and emotional responses that northern populations—with lower air conditioning prevalence and less heat-adapted infrastructure—experience immediately. Additionally, habitual indoor behavior in hot climates may reduce awareness of pollution severity until prolonged exposure or media coverage amplifies risk perception. Our study lacks data on individual mobility patterns, time spent outdoors, or air conditioning usage that would allow us to distinguish between psychological adaptation and behavioral adaptation. Future research integrating mobility data from smartphones or air conditioning penetration rates could help disentangle these behavioral and psychological pathways.

When taken together, our findings suggest that heatwaves amplify the delayed sentiment effects of air pollution in warmer regions, as illustrated by the "boiling frog effect", a metaphor describing how populations adapt to environmental stressors based on their experience, resulting in postponed emotional reactions. Our modelling results show that the expressed sentiment captured by social media users such as Twitter, Facebook, and so forth can provide a near real-time spatiotemporal indicator of how people's emotional states evolve as a function of air pollution and extreme weather. The open-access datasets and methodologies underpinning this work ensure full transparency and reproducibility. Furthermore, the framework developed in this study offers a scalable tool for tracking psychological states during emerging environmental stressors or public health crisis. By mapping the dynamic interplay between mental wellbeing and ecological stressors timely, this research advances the theoretical foundation in environmental psychology and broadens its current research paradigm. As climate change intensifies, the ability to track sentiment shifts in real-time is essential for informing public health interventions, shaping policy responses, and building climate-resilient communities.

Data Availability:

Geotweets retrieved from Harvard CGA Geotweet Archive v2.0 at https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/3NCMB6; Precipitation and heat data retrieved from NASA NLDAS-2 at https://ldas.gsfc.nasa.gov/nldas/v2/forcing; Air pollution data retrieved from USGS EROS Center at https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MCD19A2_GRANULES#bands; Other covariates retrieved from Centers for Disease Control and Prevention Social Vulnerability Index at https://www.atsdr.cdc.gov/place-health/php/svi/index.html; TIGER/Line Shapefiles retrieved from US Census Bureau at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html; Code used to retrieve air pollution data is provided through Github: https://github.com/Lilian12138/US_AOD.git

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Environmental Implications

This study demonstrates that air pollution and heatwaves jointly impose a hidden, delayed emotional burden on populations, particularly under a warming climate. The identified "boiling frog effect" suggests that adaptation to heat in warmer regions can mask the psychological impacts of worsening air quality, potentially delaying public awareness and policy response. These findings highlight the need for integrated environmental management that considers compound stressors rather than single hazards. Incorporating real-time sentiment indicators into air quality and climate monitoring systems can support earlier risk detection, more responsive environmental governance, and targeted mitigation strategies under accelerating climate change.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Highlights

- Combined heatwaves and air pollution cause delayed declines in public sentiment.
- Warmer regions show slower emotional responses, revealing the "boiling frog effect."
- Sentiment drops begin around Day 4 of pollution during heatwaves, lowest by Day 7.
- Northern states react immediately to pollution, while southern states show delay.
- Real-time tweet sentiment offers scalable tracking for climate–emotion dynamics.

Graphical abstract